

PowerShell Commands and Outputs

Microsoft Windows [Version 10.0.26200.9457]

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C:\Users\raj>mongosh

Current Mongosh Log ID: 6ac262dff882130aa1026364

Connecting to:

mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0

Using MongoDB: 9.0.2

Using Mongosh: 2.12.0

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting

2026-10-04T18:56:05.028+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

test> use collegeDB

switched to db collegeDB

collegeDB> db.students.find()

collegeDB> db

collegeDB

collegeDB> db.createCollection("students")

{ ok: 1 }

collegeDB> show collections

students

collegeDB> db.students.insertMany([

| { rollno: 101, name: "Arun", department: "CSE", year: 3, marmarks: 85 },

| { rollno: 102, name: "Priya", department: "ECE", year: 3, mamarks: 92 },

| { rollno: 103, name: "Karthik", department: "CSE", year: 2, marks: 78 },

| { rollno: 104, name: "Meena", department: "EEE", year: 3, mamarks: 88 },

| { rollno: 105, name: "Rahul", department: "CSE", year: 4, mamarks: 67 }

|])

{

acknowledged: true,

insertedIds: {

'0': ObjectId('6ac26478f882130aa1026365'),

'1': ObjectId('6ac26478f882130aa1026366'),

'2': ObjectId('6ac26478f882130aa1026367'),

'3': ObjectId('6ac26478f882130aa1026368'),

'4': ObjectId('6ac26478f882130aa1026369')

}

}

```
collegeDB> db.students.insertMany([
| { rollno: 106, name: "Divya", department: "ECE", year: 2, mamarks: 81, city: "Chennai"
| },
| { rollno: 107, name: "Sanjay", department: "CSE", year: 3, mmmarks: 95, city:
"Coimbatore" },
| { rollno: 108, name: "Anitha", department: "EEE", year: 4, mmmarks: 73, city: "Salem" },
| { rollno: 109, name: "Vijay", department: "CSE", year: 3, mamarks: 89, city: "Madurai"
| },
| { rollno: 110, name: "Nithya", department: "ECE", year: 3, mmmarks: 84, city: "Trichy" }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac2647ff882130aa102636a'),
    '1': ObjectId('6ac2647ff882130aa102636b'),
    '2': ObjectId('6ac2647ff882130aa102636c'),
    '3': ObjectId('6ac2647ff882130aa102636d'),
    '4': ObjectId('6ac2647ff882130aa102636e')
  }
}
```

```
collegeDB> db.students.find()
[
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac26478f882130aa1026366'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac26478f882130aa1026367'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
]
```

```
{
  _id: ObjectId('6ac26478f882130aa1026368'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac26478f882130aa1026369'),
  rollno: 105,
  name: 'Rahul',
  department: 'CSE',
  year: 4,
  marks: 67
},
{
  _id: ObjectId('6ac2647ff882130aa102636a'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
{
  _id: ObjectId('6ac2647ff882130aa102636b'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
  year: 3,
  marks: 95,
  city: 'Coimbatore'
},
{
  _id: ObjectId('6ac2647ff882130aa102636c'),
  rollno: 108,
  name: 'Anitha',
  department: 'EEE',
  year: 4,
  marks: 73,
  city: 'Salem'
},
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
```

```
{
  _id: ObjectId('6ac2647ff882130aa102636e'),
  rollno: 110,
  name: 'Nithya',
  department: 'ECE',
  year: 3,
  marks: 84,
  city: 'Trichy'
}
]
collegeDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac26478f882130aa1026367'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac26478f882130aa1026369'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636b'),
    rollno: 107,
    name: 'Sanjay',
    department: 'CSE',
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },
]
```

```
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
}
]
collegeDB> db.students.find({ marks: { $gt: 80 } })
[
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac26478f882130aa1026366'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac26478f882130aa1026368'),
    rollno: 104,
    name: 'Meena',
    department: 'EEE',
    year: 3,
    marks: 88
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636a'),
    rollno: 106,
    name: 'Divya',
    department: 'ECE',
    year: 2,
    marks: 81,
    city: 'Chennai'
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636b'),
    rollno: 107,
    name: 'Sanjay',
    department: 'CSE',
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },
]
```

```
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
{
  _id: ObjectId('6ac2647ff882130aa102636e'),
  rollno: 110,
  name: 'Nithya',
  department: 'ECE',
  year: 3,
  marks: 84,
  city: 'Trichy'
}
]
collegeDB> db.students.find(
| {},
| { name: 1, marks: 1, _id: 0 }
|)
[
  { name: 'Arun', marks: 85 },
  { name: 'Priya', marks: 92 },
  { name: 'Karthik', marks: 78 },
  { name: 'Meena', marks: 88 },
  { name: 'Rahul', marks: 67 },
  { name: 'Divya', marks: 81 },
  { name: 'Sanjay', marks: 95 },
  { name: 'Anitha', marks: 73 },
  { name: 'Vijay', marks: 89 },
  { name: 'Nithya', marks: 84 }
]
collegeDB> db.students.find({
| department: "CSE",
| year: 3
| })
[
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },

```

```
{
  _id: ObjectId('6ac2647ff882130aa102636b'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
  year: 3,
  marks: 95,
  city: 'Coimbatore'
},
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
}
]
collegeDB> db.students.find({
| $or: [
|   { department: "CSE" },
|   { department: "ECE" }
| ]
| })
[
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac26478f882130aa1026366'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac26478f882130aa1026367'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
]
```

```
{
  _id: ObjectId('6ac26478f882130aa1026369'),
  rollno: 105,
  name: 'Rahul',
  department: 'CSE',
  year: 4,
  marks: 67
},
{
  _id: ObjectId('6ac2647ff882130aa102636a'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
{
  _id: ObjectId('6ac2647ff882130aa102636b'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
  year: 3,
  marks: 95,
  city: 'Coimbatore'
},
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
{
  _id: ObjectId('6ac2647ff882130aa102636e'),
  rollno: 110,
  name: 'Nithya',
  department: 'ECE',
  year: 3,
  marks: 84,
  city: 'Trichy'
}
]
```

```
collegeDB> db.students.find().sort({ marks: 1 })
[
  {
    _id: ObjectId('6ac26478f882130aa1026369'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636c'),
    rollno: 108,
    name: 'Anitha',
    department: 'EEE',
    year: 4,
    marks: 73,
    city: 'Salem'
  },
  {
    _id: ObjectId('6ac26478f882130aa1026367'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636a'),
    rollno: 106,
    name: 'Divya',
    department: 'ECE',
    year: 2,
    marks: 81,
    city: 'Chennai'
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636e'),
    rollno: 110,
    name: 'Nithya',
    department: 'ECE',
    year: 3,
    marks: 84,
    city: 'Trichy'
  },
  {
    _id: ObjectId('6ac26478f882130aa1026365'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
]
```

```
{
  _id: ObjectId('6ac26478f882130aa1026368'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
{
  _id: ObjectId('6ac26478f882130aa1026366'),
  rollno: 102,
  name: 'Priya',
  department: 'ECE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac2647ff882130aa102636b'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
  year: 3,
  marks: 95,
  city: 'Coimbatore'
}
]
collegeDB> db.students.find().sort({ marks: -1 })
[
  {
    _id: ObjectId('6ac2647ff882130aa102636b'),
    rollno: 107,
    name: 'Sanjay',
    department: 'CSE',
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },

```

```
{
  _id: ObjectId('6ac26478f882130aa1026366'),
  rollno: 102,
  name: 'Priya',
  department: 'ECE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
{
  _id: ObjectId('6ac26478f882130aa1026368'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac26478f882130aa1026365'),
  rollno: 101,
  name: 'Arun',
  department: 'CSE',
  year: 3,
  marks: 85
},
{
  _id: ObjectId('6ac2647ff882130aa102636e'),
  rollno: 110,
  name: 'Nithya',
  department: 'ECE',
  year: 3,
  marks: 84,
  city: 'Trichy'
},
{
  _id: ObjectId('6ac2647ff882130aa102636a'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
```

```
{
  _id: ObjectId('6ac26478f882130aa1026367'),
  rollno: 103,
  name: 'Karthik',
  department: 'CSE',
  year: 2,
  marks: 78
},
{
  _id: ObjectId('6ac2647ff882130aa102636c'),
  rollno: 108,
  name: 'Anitha',
  department: 'EEE',
  year: 4,
  marks: 73,
  city: 'Salem'
},
{
  _id: ObjectId('6ac26478f882130aa1026369'),
  rollno: 105,
  name: 'Rahul',
  department: 'CSE',
  year: 4,
  marks: 67
}
]
collegeDB> db.students.find()
| .sort({ marks: -1 })
| .limit(3)
[
  {
    _id: ObjectId('6ac2647ff882130aa102636b'),
    rollno: 107,
    name: 'Sanjay',
    department: 'CSE',
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },
  {
    _id: ObjectId('6ac26478f882130aa1026366'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac26478f882130aa1026367'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac2647ff882130aa102636c'),
    rollno: 108,
    name: 'Anitha',
    department: 'EEE',
    year: 4,
    marks: 73,
    city: 'Salem'
  }
]
```

```

{
  _id: ObjectId('6ac2647ff882130aa102636d'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
}
]
collegeDB> db.students.countDocuments({ department: "CSE" })
5
collegeDB> db.students.updateOne(
| { rollno: 101 },
| { $set: { marks: 90 } }
|)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
collegeDB> db.students.updateMany(
| { department: "CSE" },
| { $set: { status: "Active" } }
|)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
collegeDB> db.students.deleteOne({ rollno: 105 })
{ acknowledged: true, deletedCount: 1 }
collegeDB> db.students.distinct("department")
[ 'CSE', 'ECE', 'EEE' ]
collegeDB> db.students.aggregate([
| {
|   $group: {
|     _id: "$department",
|     averageMarks: { $avg: "$marks" }
|   }
| }
| ])
[
  { _id: 'EEE', averageMarks: 80.5 },
  { _id: 'CSE', averageMarks: 88 },
  { _id: 'ECE', averageMarks: 85.66666666666667 }
]

```



